

to their existing commercial activities. Such companies are providing the government the research results the defense authorities vitally need. However, at the same time they are obtaining, at little or no cost, related non-defense research benefits which otherwise they would probably be paying for themselves. This factor may well have been one of the reasons for the spectacular investment success of Texas Instruments, Inc., which in four years rose nearly 500 per cent from the price of $5\frac{1}{4}$ at which it traded when first listed on the New York Stock Exchange in 1953; it may also have contributed, in the same period, to the even greater 700 per cent rise experienced by Ampex shareholders from the time this company's shares were first offered to the public in the same year.

Finally, in judging the relative investment value of company research organizations, another type of activity must be evaluated. This is something which ordinarily is not considered as developmental research at all—the seemingly unrelated field of market research. Market research may be regarded as the bridge between developmental research and sales. Top management must be alert against the temptation to spend significant sums on the research and development of a colorful product or process which, when perfected, has a genuine market but one too small to be profitable. By too small to be profitable I mean one that never will enjoy a large enough sales volume to get back the cost of the research, much less a worthwhile profit for the investor. A market research organization that can steer a major research effort of its company from one project which if technically successful would have barely paid for itself, to another which might cater to so much broader a market that it would pay out three times as well, would have vastly increased the value to its stockholders of that company's scientific manpower.

If quantitative measurements—such as the annual expenditures on research or the number of employees holding scientific degrees—are only a rough guide and not the final answer to whether a company has an outstanding research organization, how does the careful investor obtain this information? Once again it is surprising what the “scuttlebutt” method will produce. Until the average investor tries it, he probably will not believe how complete a picture will emerge if he asks intelligent questions about a company's research activities of a diversified group of research people, some from within the company and others engaged in related lines in competitive industries, in universities, and in government. A simpler and often